



Advanced Time Series Analysis and Forecasting

Seminar 13: Foundation models and conformal prediction



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Acronyms used in this material (1/2)

- ▣ ACI: Adaptive Conformal Inference
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ ARX: AutoRegressive model with eXogenous variables
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BG: Bulgaria (Eurostat country code)
- ▣ BH: Benjamini–Hochberg (procedure)
- ▣ CEE: Central and Eastern Europe
- ▣ CQR: Conformalized Quantile Regression
- ▣ CRPS: Continuous Ranked Probability Score
- ▣ CZ: Czechia (Eurostat country code)
- ▣ DM: Diebold–Mariano (test)
- ▣ ENTSO-E: European Network of Transmission System Operators for Electricity
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ EU: European Union
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GW: Giacomini–White (test of conditional predictive ability)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ HICP: Harmonised Index of Consumer Prices
- ▣ HU: Hungary (Eurostat country code)
- ▣ LLM: Large Language Model
- ▣ MAE: Mean Absolute Error
- ▣ PID: Proportional–Integral–Derivative (controller)
- ▣ PL: Poland (Eurostat country code)

Acronyms used in this material (2/2)

- ▣ RV: Realised Variance
- ▣ SK: Slovakia (Eurostat country code)
- ▣ VAT: Value Added Tax
- ▣ WIG20: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

Today's question and route

- **Question:** how good are zero-shot forecasts of pretrained models on our data, and how do we turn any forecast into intervals with controlled coverage when the data are dependent?
 - ▶ this seminar comes **before** Lecture 13: the section “What you need today” gives every definition the tasks use
- Route
 - ▶ Part A: split conformal and the Beta law, ACI and quantile tracking by hand, CQR and weighted quantiles, the Chronos tokeniser and its range
 - ▶ Part B: Romanian load after the model releases, covariates in context, conformal intervals for BET and DAX, calibration of Bitcoin RV forecasts, EnbPI, CEE inflation with multiple testing, VaR of the WIG20
 - ▶ Part C: a pre-registered forecast of Romanian inflation; an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

Exercise map

Task	Question	Type	Model
A1, A2	split conformal by hand; the Beta law and the calibration size	Solved, Proposed	A1
A3, A4	ACI and quantile tracking by hand; the long-run bounds	Solved, Proposed	A3
A5, A6	CQR by hand; weighted conformal quantiles	Solved, Proposed	A5
A7, A8	the Chronos tokeniser; range limits and arcsinh scaling	Solved, Proposed	A7
B1, B2	Romanian load after the releases; calendar covariates in context	Solved, Proposed	B1
B3, B4	static, rolling and adaptive conformal for BET; the ACI design on the DAX	Solved, Proposed	B3
B5, B6	online CQR for Bitcoin RV forecasts; EnbPI on the load peak	Solved, Proposed	B5
B7, B8	CEE inflation with Holm and BH; VaR of the WIG20	Solved, Proposed	B7
C1, C2	a pre-registered inflation forecast; what is wrong in an AI answer?	Proposed	B5, A1–A8

- ▣ **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency, period
Romanian electricity load	Energy-Charts (ENTSO-E)	hourly, 2022–2026
HICP annual inflation, EU countries	Eurostat	monthly, 2000–2026
Bitcoin realised variance	Binance, 5-minute returns	daily, 2018–2026
BET, DAX, WIG20	EODHD	daily, 2000–2026

- ▣ Returns are 100 times log differences; in the notebook: `fm_forecast`, `conformal_quantile`, `online_threshold`, `cqr`, `enbpi`, `dm_test`, `holm`, `bh`, `kupiec`, `christoffersen`; no account or key is needed

What you need today (1/5): foundation models

- A **foundation model** $f_{\hat{\theta}}$ is pretrained once on a corpus of many series; **zero-shot**: the forecast of a new series is $f_{\hat{\theta}}(y_{T-C+1:T})$, nothing is estimated on it (C : context length)
 - ▶ outputs: quantiles $\hat{q}_{\tau}$ at fixed levels: Chronos-Bolt, TimesFM 2.5, TiRex 10%–90% (deciles); Chronos-2 1%–99% [Ansari et al. \(2025\)](#)
- **Chronos tokeniser** [Ansari et al. \(2024\)](#): $s = \frac{1}{C} \sum_{t \leq C} |y_t|$, scaled values y_t/s , 4093 bin centres equally spaced on $[-15, 15]$; a value is replaced by the nearest centre
 - ▶ bin width 30/4092 in scaled units; values beyond 15s are clipped; Chronos-2 uses $\text{arcsinh}(y/s)$ instead
- **Covariates in context** (Chronos-2): past values of the covariates and future values of known ones (calendar) are passed with the target; their effect is inferred, not estimated
- **Contamination**: the test window may be in the training data; evaluate after the release of every model (here: from 1 November 2025)

What you need today (2/5): scores and tests

- ▣ Pinball loss $\rho_\tau(u) = u(\tau - 1\{u < 0\})$, $u = y - \hat{q}_\tau$; CRPS $\approx \frac{2}{K} \sum_k \rho_{\tau_k}(y - \hat{q}_{\tau_k})$ over K levels; MAE of the median
- ▣ Across series: ratios $r_s = L_s(\text{model})/L_s(\text{baseline})$ combined by the geometric mean $\exp(\frac{1}{S} \sum_s \ln r_s)$ [Fleming and Wallace \(1986\)](#)
- ▣ DM–HLN test ([Chapter 1](#)) on the loss differential d_t ; with S tests: **Holm Holm (1979)** controls the family-wise error, **Benjamini–Hochberg** [Benjamini and Hochberg \(1995\)](#) the false discovery rate
 - ▶ Holm: sort $p_{(1)} \leq \dots \leq p_{(S)}$, reject while $p_{(k)} \leq \alpha/(S - k + 1)$; BH: reject the k^* smallest, $k^* = \max\{k : p_{(k)} \leq k\alpha/S\}$
- ▣ Pooled test: average d_t across series at each origin, then one DM test

What you need today (3/5): split conformal and CQR

- ▣ **Exchangeable** data: the joint law is invariant to permutations (i.i.d. is a special case); a **score** $s(x, y)$, e.g. $|y - \hat{f}(x)|$
- ▣ **Split conformal** [Lei et al. \(2018\)](#): n calibration scores, $k = \lceil (n+1)(1-\alpha) \rceil$, $\hat{q} = S_{(k)}$ ($+\infty$ if $k > n$), $\hat{C}(x) = \{y : s(x, y) \leq \hat{q}\}$
 - ▶ $1 - \alpha \leq \Pr\{Y_{n+1} \in \hat{C}\} \leq 1 - \alpha + \frac{1}{n+1}$ (no ties); given the calibration set the coverage is $\text{Beta}(k, n+1-k)$
- ▣ **CQR** [Romano, Patterson and Candès \(2019\)](#): quantile forecasts $\hat{q}_{\text{lo}}, \hat{q}_{\text{hi}}$; score $\max\{\hat{q}_{\text{lo}} - y, y - \hat{q}_{\text{hi}}\}$; interval $[\hat{q}_{\text{lo}} - \hat{q}, \hat{q}_{\text{hi}} + \hat{q}]$
- ▣ **Weighted conformal** [Barber et al. \(2023\)](#): weights w_i fixed in advance, $w_{n+1} = 1$; $\hat{q}$ = the $(1-\alpha)$ quantile of $\sum_i \tilde{w}_i \delta_{S_i} + \tilde{w}_{n+1} \delta_{+\infty}$, $\tilde{w}_i = w_i / (1 + \sum_j w_j)$

What you need today (4/5): conformal methods online

- Online frame: scores S_t from out-of-sample forecasts; at time t the threshold q_t uses $S_1, \dots, S_{t-1}$ only; miss $\text{err}_t = 1\{S_t > q_t\}$
 - ▶ static: calibrated once; rolling: the conformal quantile of the last w scores
- **ACI** Gibbs and Candès (2021): q_t at level α_t , $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$; $|\frac{1}{T} \sum \text{err}_t - \alpha| \leq \frac{\max\{\alpha_1, 1-\alpha_1\} + \gamma}{\gamma T}$
 - ▶ $\alpha_t \leq 0$: $q_t = +\infty$ (an infinite interval)
- **Quantile tracking** Angelopoulos, Candès and Tibshirani (2023): $q_{t+1} = q_t + \eta(\text{err}_t - \alpha)$; if $S_t \in [0, B]$: $|\frac{1}{T} \sum (\text{err}_t - \alpha)| \leq \frac{B + \eta}{\eta T}$; PID adds an integrator and a scorecaster
- **EnbPI** Xu and Xie (2021): B models on block-bootstrap samples, leave-one-out residuals, interval $\hat{f}(x_t) \pm$ the $(1 - \alpha)$ quantile of the last residuals, updated online

What you need today (5/5): checking coverage and VaR

- ▣ Miss sequence err_t : **Kupiec** Kupiec (1995) tests $\Pr\{\text{err}_t = 1\} = \alpha$; **Christoffersen** Christoffersen (1998) adds independence (first-order Markov alternative); both χ^2 likelihood-ratio tests
- ▣ Local coverage: the share of hits over a moving window (Gibbs and Candès use 500 days); coverage by bins of a variable known at the origin
- ▣ **VaR** (Chapter 9): $\text{VaR}_\alpha = -q_\alpha$ of the return distribution; target exceedance rate α (VaR 1%: $\alpha = 0.01$); quantile loss ρ_α
 - ▶ one-sided conformal shift of a quantile forecast $\hat{q}_t$: score $\hat{q}_t - y_t$, threshold c_t at level α ; $\text{VaR} = -(\hat{q}_t - c_t)$
- ▣ GARCH(1,1)-t Bollerslev (1986) as the benchmark of Chapter 9; HAR Corsi (2009) for realised variance

A1: split conformal by hand [Solved]

Task

- ▣ 19 absolute calibration residuals, sorted: 0.2; 0.3; 0.4; 0.5; 0.6; 0.7; 0.8; 0.9; 1.0; 1.1; 1.2; 1.3; 1.5; 1.6; 1.8; 2.0; 2.1; 2.6; 3.2; a new point forecast $\hat{f} = 10$; $\alpha = 0.1$.
- ▣ 1. Compute k and $\hat{q}$.
- ▣ 2. Write the 90% interval.
- ▣ 3. Give the lower and upper bounds of the marginal coverage under exchangeability.
- ▣ Report: two numbers; one interval; two numbers and one sentence.

Solution

- ▣ 1. $k = \lceil 20 \cdot 0.9 \rceil = 18$; $\hat{q} = S_{(18)} = 2.6$
- ▣ 2. $[10 - 2.6, 10 + 2.6] = [7.4, 12.6]$
- ▣ 3. $0.90 \leq \text{coverage} \leq 0.90 + 1/20 = 0.95$
- ▣ Interpretation: with 19 points the interval uses the second-largest residual; the guarantee is over new calibration sets as well as new points

A2: the size of the calibration set [Proposed]

Solution discussed in class

Task

- ▣ $\alpha = 0.1$; given the calibration set, coverage $\sim \text{Beta}(k, n + 1 - k)$, $k = \lceil (n + 1)(1 - \alpha) \rceil$. Model: A1.
- ▣ 1. For $n = 19, 99$ and 999 give k , the mean and the standard deviation of the coverage.
- ▣ 2. For each n , compute the probability that the realised coverage is below 0.85 .
- ▣ 3. Find the smallest n whose standard deviation is at most 0.01 .
- ▣ Report: three triples; three probabilities; one number.

A3: ACI by hand [Solved]

Task

- ▣ $\alpha = 0.1$, $\gamma = 0.01$, $\alpha_1 = 0.1$; observed misses $\text{err}_1, \dots, \text{err}_5 = 1, 1, 0, 1, 0$.
- ▣ 1. Compute $\alpha_2, \dots, \alpha_6$.
- ▣ 2. Compute the bound on $|\frac{1}{T} \sum \text{err}_t - \alpha|$ for $T = 1000$.
- ▣ 3. After how many consecutive misses from α_1 is the interval infinite? After how many consecutive hits is it empty?
- ▣ Report: five numbers; one number; two integers and one sentence.

Solution

- ▣ 1. A miss subtracts $\gamma(1 - \alpha) = 0.009$, a hit adds $\gamma\alpha = 0.001$: 0.091; 0.082; 0.083; 0.074; 0.075
- ▣ 2. $(\max\{0.1, 0.9\} + 0.01)/(0.01 \cdot 1000) = 0.091$
- ▣ 3. $\lceil 0.1/0.009 \rceil = 12$ misses; $\lceil 0.9/0.001 \rceil = 901$ hits
- ▣ Interpretation: ACI reacts nine times faster to a miss than to a hit; a burst of a dozen misses makes it give up and return the whole line, which is what keeps the long-run average

A4: quantile tracking by hand [Proposed]

Solution discussed in class

Task

- ▣ Scores in $[0, 5]$; $\alpha = 0.1$, $\eta = 0.05$, $q_1 = 1$; misses 1, 0, 0, 1, 1. Model: A3.
- ▣ 1. Compute $q_2, \dots, q_6$.
- ▣ 2. Compute the bound $(B + \eta)/(\eta T)$ for $T = 2000$.
- ▣ 3. Prove that q_t stays in $[-\eta, B + \eta]$ if it starts inside $[0, B]$.
- ▣ Report: five numbers; one number; a three-line proof.

A5: CQR by hand [Solved]

Task

- ▣ 8 calibration points with quantile forecasts and outcomes $(\hat{q}_{lo}, \hat{q}_{hi}, y)$: $(1, 3, 2)$, $(2, 4, 4.6)$, $(0.5, 2.5, 0.2)$, $(1.5, 3, 2.5)$, $(3, 5, 5.4)$, $(2.5, 4, 3.1)$, $(1, 2.5, 2.2)$, $(2, 4.5, 1.6)$; $\alpha = 0.2$.
- ▣ 1. Compute the eight CQR scores.
- ▣ 2. Compute k and $\hat{q}$.
- ▣ 3. Give the conformal interval for a new point with $(\hat{q}_{lo}, \hat{q}_{hi}) = (2, 4)$ and the raw coverage on the calibration set.
- ▣ Report: eight numbers; two numbers; one interval and one share.

Solution

- ▣ 1. $-1.0; 0.6; 0.3; -0.5; 0.4; -0.6; -0.3; 0.4$
- ▣ 2. $k = \lceil 9 \cdot 0.8 \rceil = 8$, $\hat{q} = 0.6$ (the largest score)
- ▣ 3. $[2 - 0.6, 4 + 0.6] = [1.4, 4.6]$; raw coverage 0.50
- ▣ Interpretation: the raw 80% band covers only half of the points; CQR widens it by the largest miss; a negative $\hat{q}$ would narrow an over-wide band

A6: weighted conformal quantiles [Proposed]

Solution discussed in class

Task

- ▣ Six scores in time order: 2.0, 0.5, 1.5, 3.0, 1.0, 2.5; weights $w_i = 0.8^{7-i}$, $w_7 = 1$. Model: A5.
- ▣ 1. Compute the normalised weights $\tilde{w}_1, \dots, \tilde{w}_7$.
- ▣ 2. Compute the weighted and the unweighted conformal thresholds for $\alpha = 0.3$.
- ▣ 3. Repeat for $\alpha = 0.2$ and explain the result with the mass $\tilde{w}_7$.
- ▣ Report: seven weights; two thresholds; two thresholds and one sentence.

A7: the Chronos tokeniser by hand [Solved]

Task

- ▣ Context of eight values: 100, 104, 98, 102, 106, 110, 96, 104; 4093 centres equally spaced on $[-15, 15]$.
- ▣ 1. Compute the scale s and the bin width in scaled units and in data units.
- ▣ 2. Find the token of the last value (centres indexed $0, \dots, 4092$), its dequantised value and the error.
- ▣ 3. Give the largest value the model can forecast, and say what happens to a future value of 1700.
- ▣ Report: three numbers; three numbers; one number and one sentence.

Solution

- ▣ 1. $s = 102.5$; width $30/4092 = 0.00733$, i.e. 0.751 in data units
- ▣ 2. $104/s$ is closest to centre 2184; dequantised 103.70, error 0.30
- ▣ 3. $15s = 1537.5$; 1700 scales to $16.59 > 15$: clipped to the last bin, it cannot be forecast
- ▣ Interpretation: the resolution is relative to the context mean; a tenfold jump is fine, a sixteenfold one is not

A8: range limits and arcsinh scaling [Proposed]

Solution discussed in class

Task

- ▣ Context: seven values of 50 and one of 70; possible future values 90, 400, 1500. Model: A7.
- ▣ 1. Compute s and the scaled future values; say which ones Chronos cannot reach.
- ▣ 2. Compute arcsinh of the scaled values (the Chronos-2 input transform) and explain why a quantile head on this scale has no hard limit.
- ▣ 3. Name a Romanian series where this matters.
- ▣ Report: four numbers and one list; three numbers and one sentence; one example.

B1: Romanian load after the model releases [Solved]

- **Question:** is a zero-shot foundation model as accurate as the expert model for day-ahead Romanian load, on days it cannot have seen?
- **Inputs:** Energy-Charts hourly load, forecast days 1 November 2025 – 30 September 2026 (334 days); weekly naive; expert ARX of Ziel and Weron ([Chapter 1](#)); Chronos-Bolt small with 2048 hours of context
- **Tasks**
 1. Compute the MAE of the median and the CRPS over the deciles for the three models.
 2. Compute the coverage of the 10–90% band.
 3. Run the DM–HLN test of each model against the expert ARX on daily MAE.
 4. Interpretation: is the zero-shot model good enough to replace the expert model?
- **Report:** a table of three models and three scores, two DM statistics, one sentence
- **Notebook:** section B1

B1: solution [Solved]



- MAE (GW): weekly naive 0.412, expert ARX 0.227, Chronos-Bolt 0.246; CRPS 0.330, 0.184, 0.197; coverage of 80% bands 76.8%, 78.0%, 81.5%
- DM against the expert ARX: Chronos-Bolt $t = 3.06$ ($p = 0.002$); weekly naive $t = 12.69$
- Interpretation: not yet: on days it cannot have seen, Chronos-Bolt is 8% less accurate than the expert model, a significant gap, but it needs no estimation and its bands are better calibrated: a strong default, not a replacement

B2: calendar covariates in context [Proposed]

- **Question:** does Chronos-2 learn the effect of public holidays from the context alone?
- **Inputs:** the B1 window; Chronos-2 univariate and with two covariates passed for the context and for the forecast day: a holiday flag and a weekend flag; model: B1
- **Tasks**
 1. Compute the MAE of both versions on all days and on public holidays.
 2. Run the DM–HLN test of the covariate version against the univariate one.
 3. Plot the daily MAE difference and mark the holidays.
 4. Interpretation: where does the gain come from?
- **Report:** four numbers, one DM statistic, one chart, one sentence
- **Notebook:** section B2

B3: conformal intervals for BET returns [Solved]

- ▣ **Question:** does a 90% interval calibrated on a quiet period keep its coverage when volatility changes?
- ▣ **Inputs:** BET daily log returns; forecast 0, score $|r_t|$; static split calibrated on 2012–2015, rolling (last 750 days), ACI with $\gamma = 0.005$; evaluation 2016–2026 ($n = 2675$)
- ▣ **Tasks**
 1. Compute the overall coverage of the three methods and the coverage by year.
 2. Run the Kupiec and Christoffersen tests on the miss sequences.
 3. Find the worst year of the static method.
 4. Interpretation: which property of returns breaks the static method?
- ▣ **Report:** three coverages, six p -values, one year, one sentence
- ▣ **Notebook:** section B3

B3: solution [Solved]



- Coverage: static 0.876, rolling 0.890, ACI 0.897; Kupiec $p < 0.001$, 0.081, 0.586; Christoffersen $p < 0.001$, < 0.001 , < 0.001
- Static method by year: from 0.765 (2026) to 0.956; its threshold is fixed at 1.29%
- Interpretation: volatility clustering: the score $|r_t|$ is not exchangeable, so a quiet calibration under-covers in turbulent years; adaptive methods repair the frequency, but misses still cluster (Christoffersen)

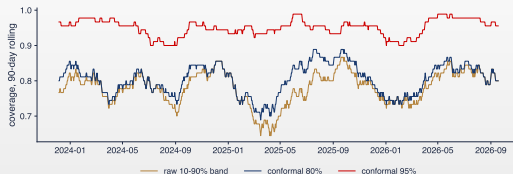
B4: the ACI volatility design on the DAX [Proposed]

- ▣ **Question:** does the result of Gibbs and Candès (2021) hold for the DAX, and how does it depend on γ ?
- ▣ **Inputs:** DAX daily returns; $V_t = r_t^2$, GARCH(1,1) on the previous 1250 days, score $|V_t - \hat{\sigma}_t^2|/\hat{\sigma}_t^2$, $\alpha = 0.1$; model: B3
- ▣ **Tasks**
 1. Compute the rolling (fixed-level) intervals and the ACI intervals for $\gamma = 0.001, 0.005$ and 0.02 .
 2. Report the overall coverage, the range of the 500-day local coverage and the share of infinite intervals.
 3. Run the Christoffersen test for each method.
 4. Interpretation: which γ would you choose, and why?
- ▣ **Report:** one chart, four triples, four p -values, one sentence
- ▣ **Notebook:** section B4

B5: calibrating a foundation model on Bitcoin RV [Solved]

- ▣ **Question:** are the 10–90% bands of Chronos-Bolt for Bitcoin log realised variance calibrated, and can conformal prediction extend them to 95%?
- ▣ **Inputs:** Binance 5-minute RV, log scale, one day ahead from 2023, context 1024 days; online CQR: score $\max\{\hat{q}_{0.1} - y, y - \hat{q}_{0.9}\}$, ACI on the last 250 scores, $\gamma = 0.005$
- ▣ **Tasks**
 1. Compute the coverage and the mean width of the raw band.
 2. Compute the coverage and the mean width of the conformal 80% and 95% intervals.
 3. Plot the 90-day rolling coverage of the three intervals.
 4. Interpretation: what does the conformal correction say about the model?
- ▣ **Report:** two numbers, four numbers, one chart, one sentence
- ▣ **Notebook:** section B5

B5: solution [Solved]



- Raw band: coverage 0.780, width 1.31; conformal 80%: 0.802, width 1.38; conformal 95%: 0.950, width 2.32 (log units, 1107 days from 8 September 2023)
- Interpretation: the deciles of a zero-shot model are close to calibrated on Bitcoin RV (78% for 80%); the conformal step adds the missing points and builds a 95% interval, 1.78 times the raw width, that the model cannot output

B6: EnbPI on the evening load peak [Proposed]

- **Question:** does a bootstrap ensemble with a sliding residual window give valid and sharp intervals for the 19:00 load?
- **Inputs:** Romanian load at 19:00, 2025–2026; EnbPI: ridge regressions on the expert ARX regressors, 20 block-bootstrap samples (blocks of 7 days) of the previous year; ACI around the expert ARX forecast (91-day window); model: B5
- **Tasks**
 1. Compute the 90% EnbPI intervals and the ACI intervals.
 2. Report coverage, mean width and the Christoffersen p -value of each.
 3. Plot both intervals and the load.
 4. Interpretation: which method would a grid operator prefer?
- **Report:** six numbers, one chart, one sentence
- **Notebook:** section B6

B7: CEE inflation, many countries, many tests [Solved]

- **Question:** do Chronos-2 and TimesFM beat AR for CEE inflation six months ahead, after correcting for multiple tests?
- **Inputs:** Eurostat HICP annual inflation, BG, CZ, HU, PL, RO, SK; monthly origins January 2012 – February 2026 (170 origins); $h = 6$; $AR(p)$ by AIC; foundation models on the full history
- **Tasks**
 1. Compute the MAE ratio to AR by country and its geometric mean, for both models.
 2. Run the six country DM–HLN tests (HAC with 5 lags) and count rejections at 5%: raw, Holm, BH.
 3. Run the pooled DM test on the cross-country average loss differential.
 4. Interpretation: what is the honest one-line summary?
- **Report:** twelve ratios, two geometric means, six counts, two pooled tests, one sentence
- **Notebook:** section B7

B7: solution [Solved]



- Chronos-2 / AR: BG 1.03; CZ 0.91; HU 0.87; PL 0.87; RO 1.13; SK 0.92; geometric mean 0.952; rejections raw/Holm/BH: 1/0/0; pooled $t = -1.10$ ($p = 0.273$)
- TimesFM / AR: BG 1.02; CZ 0.81; HU 0.93; PL 0.93; RO 1.05; SK 0.92; geometric mean 0.938; rejections 0/0/0; pooled $t = -1.12$ ($p = 0.264$)
- Interpretation: both geometric means are below one, but no country difference survives Holm or BH and neither pooled test is significant: a modest average gain that these 14 years of six countries cannot confirm; Romania is where Chronos-2 does worst

B8: VaR of the WIG20 from foundation models [Proposed]

- **Question:** can a foundation model, alone or calibrated, deliver a VaR that passes the backtests of [Chapter 9](#)?
- **Inputs:** WIG20 daily returns, 2016–2026; GARCH(1,1)- t (1000 days, re-estimated every 250); Chronos-2 raw quantiles and with ACI; Chronos-Bolt 10% quantile shifted by a one-sided conformal threshold (ACI, window 500); model: B3 and B5
- **Tasks**
 1. Compute VaR 1% and VaR 5% with the four methods.
 2. Report hit rates, Kupiec and Christoffersen p -values.
 3. Compare the quantile losses with GARCH- t by a DM test.
 4. Interpretation: what does calibration fix, and what does it not fix?
- **Report:** a table of eight rows, one sentence
- **Notebook:** section B8

C1: a pre-registered inflation forecast [Proposed]

- **Question:** what will Romanian HICP inflation be in each of the next six months, with a calibrated 80% band, written down before the data exist?
- **Inputs:** Eurostat HICP, last observation August 2026 (6.3%); Chronos-2 medians and 10–90% bands from the last origin, widened horizon by horizon by the conformal threshold of the CQR scores of the last 60 origins; model: B5
- **Tasks**
 1. Write a five-line pre-registration: data vintage, model, horizons, scoring rule, what would count as a failure.
 2. Compute the six medians and the raw and conformal 80% bands.
 3. Add one covariate you believe matters and say how you would pass it to Chronos-2.
 4. Interpretation: how will you know in six months whether the forecast was good?
- **Report:** a five-line plan, a table of six months, a project plan
- **Notebook:** section C1

C2: audit an AI answer [Proposed]

- ▣ A student asked an AI assistant to summarise foundation models and conformal prediction. The answer:
- ▣ (a) Chronos-Bolt gives a VaR 1% directly: request the 0.01 quantile.
- ▣ (b) Split conformal guarantees 90% coverage for every value of the covariate.
- ▣ (c) ACI guarantees that misses are independent over time.
- ▣ (d) With 19 calibration scores and $\alpha = 0.1$, the threshold is the 18th smallest score.
- ▣ (e) Any stationary time series is exchangeable, so split conformal is exact for it.
- ▣ (f) The Holm correction controls the false discovery rate.
- ▣ (g) The geometric mean of relative errors ranks models the same way whatever baseline is used.
- ▣ (h) Tan et al. (2024) showed that time-series foundation models such as Chronos are not useful.
- ▣ **Tasks**
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (notebook section C2).
 - ▶ 2. Report: a list of eight verdicts with one line of justification each.

Key takeaways

- ▣ A conformal threshold is an order statistic: its coverage is exact on average and Beta-distributed given the calibration set
- ▣ Under dependence, feedback (ACI, quantile tracking) restores the long-run frequency for any sequence, not the independence of misses
- ▣ Foundation-model quantiles are scores to be calibrated; tokenisers and quantile heads limit what they can say
- ▣ Many series mean many tests: pool or correct before declaring a winner
- ▣ An AI answer is a draft: check quantile ranges, guarantees and the scope of cited studies

After the seminar

- Lecture 13 develops each topic of today: pretraining and model design, benchmarks and contamination, conformal theory and its time-series versions, calibration of foundation models
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: a live, pre-registered benchmark of foundation models on CEE inflation with conformal bands
- Reading: Angelopoulos and Bates (2023); Gibbs and Candès (2021); Romano, Patterson and Candès (2019); Ansari et al. (2024); Tan et al. (2024)
- **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

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